

Alhassan Jawad

Electrical Engineer | Focusing on OT-Security

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Languages: Swedish (Fluent), English (Fluent), Arabic (Native)

Profile & Engineering Focus

Electrical Engineer (B.Sc.) currently working part-time at Abbott Laboratories while completing an M.Sc. in Electrical Engineering with a specialization in **Control of Electrical Systems**. My engineering background connects physical control theory, embedded firmware development, and industrial infrastructure administration. Practical experience includes low-level C/C++ development on dual-core microcontrollers, bare-metal hardware interfacing (ATmega328P/ESP32), active identity management in enterprise environments (RBAC). Looking ahead, I am focusing my engineering career on OT and ICS cybersecurity, building on graduate coursework in power grid security to evaluate and protect industrial control architectures using frameworks such as NIST SP 800-82 and ISO/IEC 27001.

Education & Academic Background

Uppsala University — Master of Science (M.Sc.) in Electrical Engineering
2025 – 2027

Specialization: Control of Electrical Systems (Styrning av elektriska system)

Key Courses: Cyber Security in Power Systems (5 ECTS), Automatic Control I-III (15 ECTS), Sensors for Electric Vehicles (5 ECTS), Automation & Robotics (5 ECTS), Rotating Electrical Machines (5 ECTS), High-Performance & Parallel Computing (10 ECTS), PCB Design with ECAD Tools (10 ECTS).

Uppsala University — Bachelor of Science (B.Sc.) in Electrical Engineering
2019 – 2022

Department of Electrical Engineering

Core coursework: Linear dynamic systems, feedback control loops, electric circuits, power electronics, and analog/digital signal conditioning.

Certifications

- **micro1:** Vetted Software & Electronic Engineer in Ngspice circuit simulation (Issued 2026)
- **Google Cloud Certified:** Professional Cloud AI Engineer (Issued 2026)

Technical Competencies & Systems

- **OT/ICS & Infrastructure Security:** NIST SP 800-82, ISO/IEC 27001, SCADA/ICS security architectures, Active Directory (RBAC/LDAP), network segmentation (VLANs), TCP/IP networking, Windows Server, Linux environments.
- **Control Systems & Power Engineering:** Multivariable control (MIMO), state-space modeling, feedback control loops, Hardware-in-the-Loop (HIL) simulation, power load-shift optimization.
- **Embedded Firmware & Hardware:** C, C++, Embedded C, microcontrollers (ESP32, ATmega328P), bare-metal register configuration, hardware timer interrupts, communication protocols (I2C, UART, SPI), Altium Designer (multi-layer PCB design).
- **Programming & Computational Tools:** Python, Java, MATLAB/Simulink, PowerShell, Bash, SQL (PostgreSQL), Git version control, LabVIEW, LTSpice, Ngspice.

Work Experience

Abbott Laboratories – IT Infrastructure Specialist
Sep 2026 – Present | Solna, Sweden

BTS Facility Infrastructure Operations

- Resolve complex hardware and facility network infrastructure faults across Nordic sites (Sweden, Denmark, Finland), serving as Tier 3 technical escalation for critical system availability.
- Enforce Role-Based Access Control (RBAC) and least-privilege principles within Active Directory, auditing permission groups to isolate administrative privileges across enterprise systems.
- Configure and stage network switches, routers, and secure subnets (VLANs) to maintain local network segmentation and operational uptime across facility environments.
- Develop and execute PowerShell automation scripts to standardize operating system configurations, automate endpoint compliance checks, and streamline deployment workflows.

Work Experience (Continued)

Portabel Health – Software Engineer Intern **Sep 2025 – Dec 2025 | Remote**

- Managed backend relational schemas and optimized SQL queries in PostgreSQL, ensuring data consistency, indexing efficiency, and reliable time-series data storage.
- Standardized Git branching protocols and change documentation, establishing strict codebase traceability and auditability across development cycles.
- Executed functional test suites and verification procedures across development and pre-production environments to identify regressions and validate backend data pipelines.

Uppsala University – Teaching Assistant **Aug 2023 – Jul 2025 | Uppsala, Sweden**

Electrical Engineering & Programming (Department of Information Technology)

- Instructed laboratory sessions and evaluated technical coursework for 50+ engineering students per semester across circuit analysis, embedded C/Python programming, and physical instrumentation.
- Conducted hands-on debugging sessions utilizing mixed-signal oscilloscopes, multimeters, and arbitrary waveform generators while strictly enforcing high-voltage laboratory safety protocols.
- Calibrated and verified physical measurement test benches in collaboration with faculty to ensure signal integrity and accurate baseline telemetry across experimental hardware.

Selected Engineering Projects

Master's Thesis: Automated Aeroponic Cultivation System **2025 | Uppsala University**

- Engineered a closed-loop MIMO feedback control algorithm in Embedded C on an ESP32 dual-core microcontroller to regulate dynamic environmental setpoints in real time.
- Configured a non-blocking I2C telemetry state machine to poll multi-sensor digital arrays without bus starvation, feeding real-time telemetry directly into actuation routines.
- Authored direct register configurations, hardware timer ISRs, and startup routines to achieve deterministic execution and microsecond-level timing across sensor acquisition cycles.

Hardware-in-the-Loop Elevator Prototype **2023 | Uppsala University**

Project in Electronics, 10 ECTS

- Constructed and verified a physical three-level elevator prototype using a Hardware-in-the-Loop (HIL) testbench to validate closed-loop control routines.
- Designed deterministic finite-state machines (FSM) and motor velocity profiles in LabVIEW, tuning acceleration curves to dampen mechanical resonance and eliminate jerk.
- Implemented physical fail-safe logic and hardware limit switches, establishing boundary safety trip conditions for fault mitigation during anomalous operating states.

Bachelor's Thesis: Smart Building Energy Management **2022 | Studenternas IP, Uppsala**

External Collaboration with Studenternas IP, Uppsala

- Developed a numerical optimization framework in MATLAB to evaluate grid power demand profiles and dynamic load distribution for a commercial sports facility.
- Processed high-resolution time-series meter telemetry, isolating 71.2 kW of a 185 kW peak demand as controllable flexibility for peak-shaving routines.
- Demonstrated a 38.5% reduction in peak grid electrical demand through demand-side load shifting, mitigating tariff spikes while maintaining operational constraints.

Multi-Layer Telemetry Unit & Custom PCB **2022 | Uppsala University**

PC-Board Construction with ECAD Tools, 10 ECTS

- Designed a multi-layer PCB in Altium Designer, routing for trace clearance, high-frequency signal integrity, and analog/digital domain isolation.
- Developed bare-metal C firmware on an ATmega328P, interfacing via hardware UART to a Quectel MC60 GSM/GPS module to parse NMEA sentences into formatted TCP telemetry payloads.
- Implemented low-power sleep routines using external hardware interrupts driven by an MMA8653 3-axis accelerometer for event-triggered telemetry wake-up.

References

References available upon request.